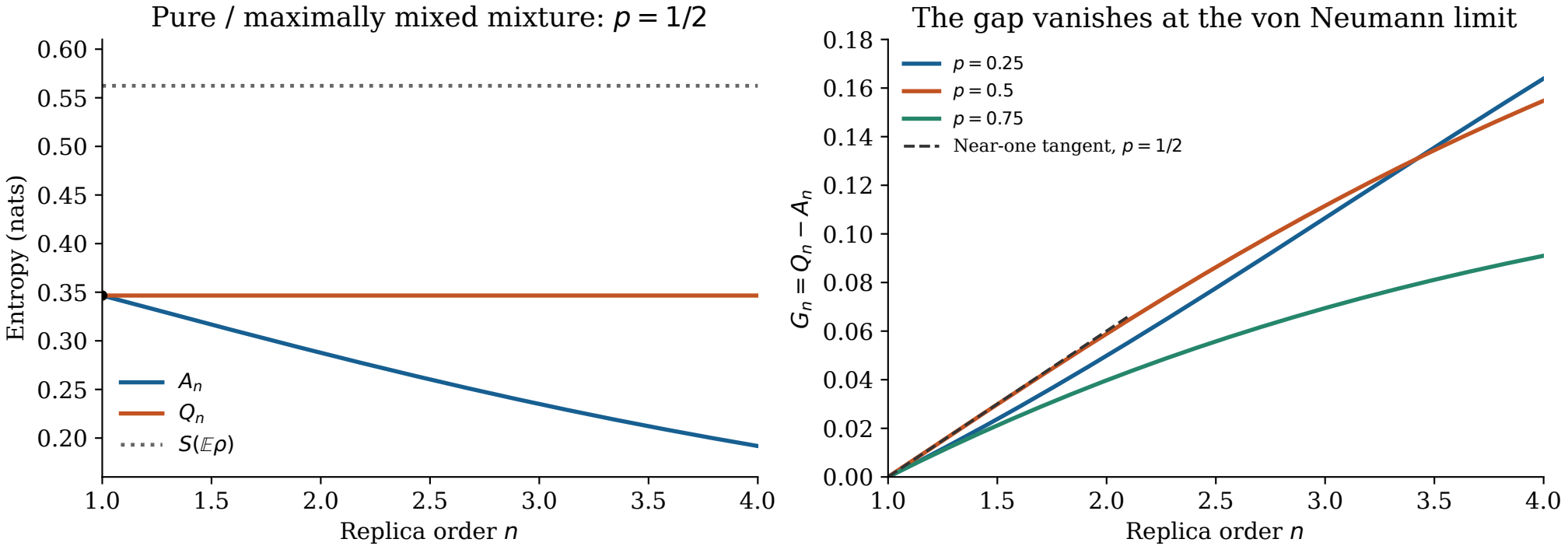
Pure / maximally mixed mixture: $p = 1/2$
Entropy (nats)
0.60
0.55
0.50
0.45
0.40
0.35
0.30
0.25
0.20
A_n
Q_n
$S(\mathbb{E}\rho)$
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Replica order n
The gap vanishes at the von Neumann limit
$G_n = Q_n - A_n$
0.18
0.16
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
$p = 0.25$
$p = 0.5$
$p = 0.75$
Near-one tangent, $p = 1/2$
1.0
1.5
2.0
2.5
3.0
3.5
4.0
Replica order n